WILP, BITS Pilani

Session 14: Advanced Time Series (ARIMA, SARIMAX, VAR)

Course: Introduction to Statistical Methods (ISM) | Instructor: Dr. YVK Ravi Kumar

1 White Noise, Dickey-Fuller Test & ARMA Models

Diagnose white noise residuals, test stationarity, and combine past memory with shocks.

A time series is white noise if $\epsilon_t \sim \text{i.i.d.}(0, \sigma^2)$ with zero autocovariance. The Dickey-Fuller test checks for a unit root ($\phi = 1$) in $\Delta y_t = (\phi - 1)y_{t-1} + \epsilon_t$. The ARMA$(p, q)$ model combines autoregressive memory with moving average shock corrections:

$$y_t = c + \phi_1 y_{t-1} + \cdots + \phi_p y_{t-p} + \epsilon_t + \theta_1 \epsilon_{t-1} + \cdots + \theta_q \epsilon_{t-q} \quad (1)$$

Everyday Picture & Intuition

White noise is pure static on a radio. The Dickey-Fuller test checks whether your radio station drift is stationary, while ARMA models memory plus lingering surprise shocks.

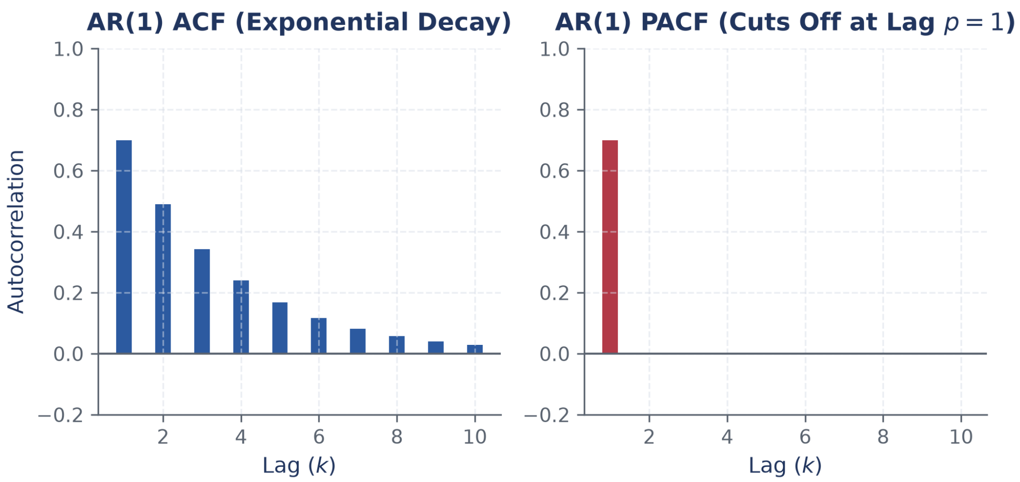

Figure 1: Autocorrelation Function (ACF) exponential decay vs Partial Autocorrelation (PACF) cutoff.

Fully Worked Example

Given an ARMA(1,1) model: $y_t = 3 + 0.5y_{t-1} + \epsilon_t + 0.4\epsilon_{t-1}$. Last observed value $y_{100} = 10$, and last residual $\epsilon_{100} = 2$.

Step 1: Forecast 1 step ahead ($t = 101$):

$$\hat{y}_{101} = 3 + 0.5(10) + 0.4(2) = 3 + 5 + 0.8 = 8.8 \tag{2}$$

Step 2: Forecast 2 steps ahead ($t = 102$): Since ϵ_{101} is unknown, its expected value $E[\epsilon_{101}] = 0$. The MA term drops out:

$$\hat{y}_{102} = 3 + 0.5(8.8) = 3 + 4.4 = 7.4 \tag{3}$$

Notice how the MA shock term only affects the immediate 1-step forecast!

2 ARIMA & SARIMAX: Differencing, Seasonality, and External Factors

Stationarize trending data via differencing and layer seasonal and exogenous terms.

ARIMA(p, d, q) differences non-stationary series d times ($w_t = \Delta^d y_t$) before fitting ARMA. SARIMAX extends this by adding seasonal periods m and exogenous regressors X_t:

$$\Phi_P(B^m)\phi_p(B)\Delta^d\Delta_m^D y_t = \beta X_t + \Theta_Q(B^m)\theta_q(B)\epsilon_t \tag{4}$$

Everyday Picture & Intuition

ARIMA strips away constant upward trends. SARIMAX adds seasonal calendars (e.g. December holiday surges) and external drivers like promotional marketing discounts.

Fully Worked Example

Consider an ARIMA(1,1,1) model fitted on differenced series $w_t = y_t - y_{t-1}$: Fitted equation: $w_t = 0.4w_{t-1} + \epsilon_t + 0.3\epsilon_{t-1}$. Given $y_{99} = 115, y_{100} = 120 \implies w_{100} = 5$. Last error $\epsilon_{100} = 1$.

Step 1: Forecast differenced value $\hat{w}_{101}$:

$$\hat{w}_{101} = 0.4(5) + 0.3(1) = 2.0 + 0.3 = 2.3 \tag{5}$$

Step 2: Un-difference to get level forecast $\hat{y}_{101}$:

$$\hat{y}_{101} = y_{100} + \hat{w}_{101} = 120 + 2.3 = 122.3 \tag{6}$$

Step 3: Forecast step 2 differenced value $\hat{w}_{102}$ ($E[\epsilon_{101}] = 0$):

$$\hat{w}_{102} = 0.4(2.3) = 0.92 \implies \hat{y}_{102} = 122.3 + 0.92 = 123.22 \tag{7}$$

3 Vector Autoregression (VAR & VARMAX)

Model multiple interdependent time series with mutual cross-lag feedback.

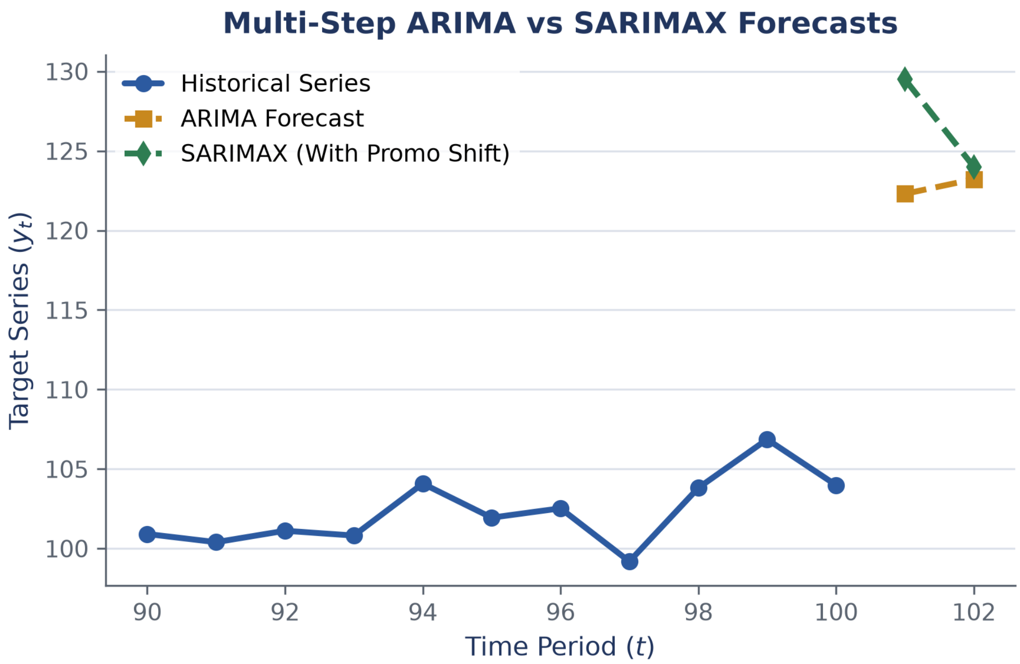

Figure 2: Multi-step forecasts comparing ARIMA baseline vs SARIMAX promotion-adjusted forecasts.

VAR(p) models a vector of time series $\mathbf{Y}_t$, where each series depends on past values of itself and past values of every other series in the system:

$$\mathbf{Y}_t = \mathbf{c} + \mathbf{A}_1\mathbf{Y}_{t-1} + \cdots + \mathbf{A}_p\mathbf{Y}_{t-p} + \boldsymbol{\epsilon}_t \tag{8}$$

Everyday Picture & Intuition

GDP growth affects inflation, but inflation also feeds back into GDP growth. VAR treats both as equal partners in a joint dynamic ecosystem.

Fully Worked Example

Given a 2-variable VAR(1) system for GDP growth (g_t) and Inflation (π_t):

$$g_t = 0.3 + 0.5g_{t-1} + 0.1\pi_{t-1}$$
$$\pi_t = 0.2 + 0.2g_{t-1} + 0.6\pi_{t-1}$$

Given last quarter values $g_{99} = 2.0\%$ and $\pi_{99} = 3.0\%$.

Step 1: Forecast GDP growth for quarter 100:

$$\hat{g}_{100} = 0.3 + 0.5(2.0) + 0.1(3.0) = 0.3 + 1.0 + 0.3 = 1.6\% \tag{9}$$

Step 2: Forecast Inflation for quarter 100:

$$\hat{\pi}_{100} = 0.2 + 0.2(2.0) + 0.6(3.0) = 0.2 + 0.4 + 1.8 = 2.4\% \tag{10}$$

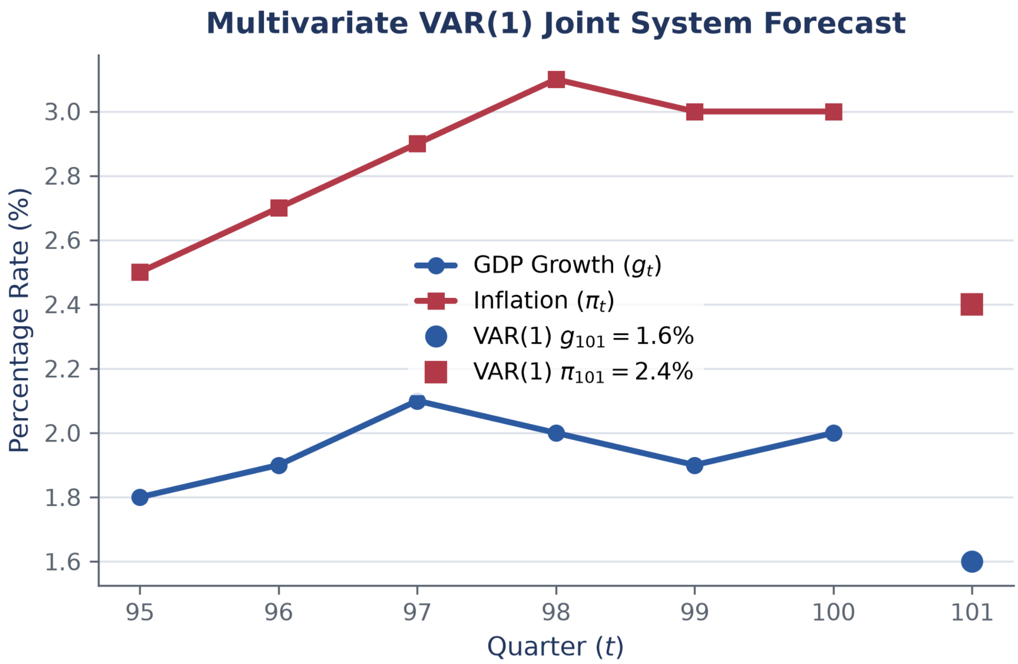

Figure 3: Multivariate VAR(1) joint forecast tracking interdependencies between GDP growth and inflation.

Watch Out! Pitfalls

VAR models require all target series to be stationary. Use the Dickey-Fuller test to verify stationarity before fitting VAR.

Key Takeaways

- White Noise residuals indicate well-fitted model dynamics.
- Dickey-Fuller tests verify stationarity.
- ARMA combines past memory (AR) and shock residuals (MA).
- ARIMA uses differencing d to stationarize trends; SARIMAX adds seasonal periods m and external regressors X.
- VAR models multivariate time series systems with mutual feedback across variables.